Heterogeneous Verification for Multi-Step Decentralized Workflows

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Abstract

Verifiable multi-step automation would extend blockchain guarantees beyond single-transaction execution, enabling temporal workflows with cryptographic attestations. Smart contracts provide verification for onchain computation, but these guarantees are lost when workflows require offchain execution such as external API calls, confidential data, or non-deterministic operations. We propose a heterogeneous verification system where each workflow step employs the verification mechanism appropriate to its computation type. The system composes cryptographic proofs for blockchain operations, hardware attestations for confidential execution, shadow computation for AI inference, and economic security for external interactions into unified workflow attestations. Each step produces a signed attestation that validators broadcast and stake against. As long as honest validators control sufficient stake, fraudulent attestations can be challenged through onchain fraud proofs, maintaining verifiability across heterogeneous trust assumptions. The system enables verifiable automation previously impossible under uniform trust models.

1 Introduction

References

[1] S. Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System," 2008.